Md Kamrul Islam

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Google Scholar | ORCID | Portfolio | GitHub | LinkedIn

Research Interests

LLMs, Retrieval-Augmented Generation, Agentic AI, Multimodal AI, Information Retrieval, and Efficient AI

Academic & Research Experience

Lecturer [Faculty Profile] *April 2026 – Present*
Department of Computer Science, American International University-Bangladesh (AIUB) *Dhaka, Bangladesh*

- Coordinate courses in Data Processing and Visualization, Programming for Data Science, and Compiler Design.
- Develop course plans and materials; mentor students and supervise undergraduate theses.

Tech Stack: *Latex, MS Tools, Outcome-Based Education (OBE)*

AI Research Engineer Intern [Preprint] *May 2025 – Oct 2025*
LISSI, Université Paris-Est Créteil | in collaboration with CEA–LIST *Paris, France*

- Developed a hybrid LLM-RAG framework for security annotation generation from natural language.
- Built a benchmark of 27 SecBPMN-text pairs to evaluate the framework against domain experts.
- Achieved 0.58 precision and 0.52 recall, reducing annotation time by 95%.

Tech Stack: *LLMs, RAG, Python, LangChain*

Graduate Research Assistant [Code] *Oct 2024 – Mar 2025*
Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), CentraleSupélec *Paris, France*

- Developed a group-equivariant CNN pipeline for chest X-ray clustering.
- Achieved faster convergence and improved accuracy over the baseline.
- Demonstrated reduced reliance on data augmentation through geometric equivariance.

Tech Stack: *PyTorch, OpenCV, CUDA, HPC*

Education

Erasmus Mundus Joint Master's Degree in Big Data Management and Analytics (BDMA) *Sept 2023 – Oct 2025*
CentraleSupélec | Universitat Politècnica de Catalunya | Université libre de Bruxelles *France | Spain | Belgium*

- **Grade:** *A (French distinction: Mention Bien)*
- **Thesis:** *Automated Generation of Cybersecurity-Aware Business Process Models using Large Language Models*

Bachelor of Engineering in Software Engineering *Mar 2018 – Dec 2021*
Sichuan University *Chengdu, China*

- **CGPA:** 3.64/4.00 (87%, Top 5%)
- **Thesis:** *Brain Tumor Detection and Classification Using Convolutional Neural Networks*

Publications

1. **M. K. Islam**, T. Henry, M. Salnitri, J. Köpke, and S. Souihi, "A Hybrid LLM-Based Framework for Automated Security Annotation Generation in Business Process Models," in *Proceedings of the 24th International Conference on Business Process Management (BPM 2026)*, Toronto, ON, Canada, 2026, pp. 168–184. *Lecture Notes in Business Information Processing*, vol. 599, Springer, Cham. [DOI]
2. C. K. Sah, L. Xiaoli, M. M. Islam, and **M. K. Islam**, "Navigating the AI Frontier: A Critical Literature Review on Integrating Artificial Intelligence into Software Engineering Education," in *Proceedings of the 36th International Conference on Software Engineering Education and Training (CSEE&T 2024)*, Würzburg, Germany, 2024, pp. 1–5. [DOI]

Workshops

M. K. Islam, X. Zhuang, J. Ma, and A. Malhotra, "Unsupervised Chest X-ray Clustering with Group Equivariant Convolutions," *RAMH 2026: Recent Advances in Machine Learning for Healthcare*, Paris, France. [HAL Record] [Poster]

Professional Experience

Software Engineer Intern *Dec 2020 – May 2021*
Chengdu Suncape Co., Ltd. *Chengdu, China*

- Built distributed data pipelines for large-scale ingestion, transformation, and analytics.
- Optimized preprocessing and feature engineering, improving model accuracy by 20%.
- Implemented modular software and CI/CD practices in an Agile Scrum team.

Tech Stack: *Apache Spark, Scikit-learn, SQL, CI/CD*

Technical Skills

Programming Languages: Python, C/C++, SQL

ML Frameworks & Libraries: Scikit-learn, PyTorch, OpenCV

LLMs & Agentic AI: LLMs, RAG, LangChain, LangGraph

Data Engineering & Knowledge Graphs: Apache Spark, PostgreSQL, Neo4j, SPARQL

Tools & Computing: Git, CUDA, HPC

Projects

DigiScan360 [Code] *Feb 2024 – June 2024*
Developed an LLM-powered competitive intelligence prototype using big data and knowledge graphs.
Tech Stack: *PySpark, LLMs, Microsoft Fabric, Power BI, GraphDB, SPARQL*

Anomaly Detection in Diesel Train Cooling Systems [Code] *Sept 2023 – Dec 2023*
Developed unsupervised models to detect anomalies from spatio-temporal data in train cooling systems.
Tech Stack: *Python, SQL, Scikit-learn, Tableau*

Brain Tumor Detection and Classification Using Convolutional Neural Networks [Code] *Sept 2021 – Dec 2021*
Achieved 98% accuracy in multi-class tumor classification using a custom CNN model.
Tech Stack: *Python, TensorFlow, OpenCV*

Academic Service & Activities

Peer Reviewer – 4th International Conference on Computing Advancements, 2026 [Certificate] *Aug 2026*
ACM SIGAPP In-Cooperation | American International University-Bangladesh *Dhaka, Bangladesh*

Twelfth European Big Data Management & Analytics Summer School (eBISS 2024) [Poster] *July 2024*
University of Padova *Padova, Italy*

Honors & Scholarships

Erasmus Mundus Scholarship *2023 – 2025*
European Commission – Fully funded merit-based scholarship for the BDMA Joint Master's Programme

Belt and Road Initiative Scholarship *2018 – 2022*
Chinese Government – Scholarship worth CNY 80,000 over four years for undergraduate study

Languages

English: Proficient | **Chinese:** Upper-intermediate | **French:** Elementary | **Bengali:** Native

References

References available upon request.